

Trigonometric Equations (Homework)

Name
Date
Course

* When solving the following problems, do not use a calculator if your answer(s) are 0° , multiples of 30° , or multiples of 45° .

① Solve if $0 \leq \theta \leq 360^\circ$. ② Find all values of α .

$$2\sqrt{3}\cot\theta + 2 = 0$$

$$4 = 3 + \sqrt{2}\cos\alpha$$

③ Solve if $0 < B < 360^\circ$.

$$-7\sin B + 1 = -3\sin B - 3$$

④ Solve if $0 < \beta \leq 360^\circ$.

* Use a calculator.

$$4 + \tan\beta = 13\tan\beta + 40$$

⑤ Solve if $0 < A < 360^\circ$.

* Use a calculator.

$$7\sin A = 3\csc A$$

⑥ Solve if $0 \leq W < 360^\circ$.

$$\tan W = \cot W$$

⑦ Solve if $0 < \theta < 360^\circ$.

$$\cos\theta - \sec\theta = 0$$

⑧ Solve if $0 \leq X \leq 360^\circ$.

$$\frac{5}{\sin X} - 4\csc X = 2$$

⑨ Solve if $0 \leq B \leq 360^\circ$ ⑩ Solve if $0 \leq y \leq 360^\circ$

$$\tan B = \sin B$$

$$\sqrt{3} \cos y = \sin y$$

⑪ Find all values of w . ⑫ Solve if $0 < \theta < 360^\circ$

$$\csc^2 w + 1 = \cot^2 w + \csc w$$

$$\sin^2 \theta - 1 = \cos \theta$$

⑬ Solve if $0 < A \leq 360^\circ$

* Hint: Square both sides of the equation.

$$\tan A + 1 = \sec A$$

Homework Answers

① $\theta = 120^\circ, 300^\circ$

⑫ $\theta = 90^\circ, 180^\circ, 270^\circ$

② $\alpha = 45 + 360n$, n is any integer.
 $\alpha = 315 + 360n$, n is any integer.

⑬ $A \neq 180^\circ$
 $A = 360^\circ$

③ $\beta = 90^\circ$

④ $\beta = 108.4^\circ, 288.4^\circ$

⑤ $A \approx 40.89^\circ, 139.1^\circ, 220.89^\circ, 319.1^\circ$

⑥ $W = 45^\circ, 135^\circ, 225^\circ, 315^\circ$

⑦ $\theta = 180^\circ$

⑧ $X = 30^\circ, 150^\circ$

⑨ $\beta = 0^\circ, 180^\circ, 360^\circ$

⑩ $y = 60^\circ, 240^\circ$

⑪ $W = 30 + 360n$, n is any integer
 $W = 150 + 360n$, n is any integer